Robotic Ultrasound Arm

Design review and calculation report — verification of the existing four-actuator arm (EIR report and CAD drawings), corrected joint-torque sizing on the final geometry, the additional checks needed before fabrication, and what autonomous scanning adds

Prepared for	Amit Kumar, B.Tech Mechanical Engineering, IIT Guwahati — BTP: robotic ultrasound scanning system (guide: Prof. S. K. Dwivedy)
Documents reviewed	EIR report "Teleoperated Ultrasound System" (16 pp., scanned) · Robotic arm assembly (A2, 16-08-2023) · Robotic arm drawings, sheets 1–11 (28-08-2023) · Robotic arm exploded view (D, 16-08-2023)
Companion document	Component list, analysis and BTP feasibility (second PDF)
Date	13 October 2026 — revision B: joint spacings measured on the final assembly drawing (379 / 372 mm); checks for fully autonomous scanning added (Section 7)

Summary of findings

#	Check	Result	Verdict
1	Arithmetic of the original sizing	16.47 N·m reproduces, but the 1.5 factor is applied twice to the acceleration term and the load inputs (J_L, T_L) cannot be traced to the stated masses	▲ Partly valid
2	Servo chosen vs the report's own requirement	80 kgf·cm (7.85 N·m) servo against 16.47 N·m required	✗ Inconsistent
3	Axis 2 (shoulder) holding torque	13.6 N·m in the drawn pose, 25.0 N·m with links horizontal; motor 12 N·m holding, ≈8.4 N·m usable	✗ Not OK
4	Axis 3 (elbow) holding torque	10.0 N·m drawn pose, 9.8–10.3 N·m over the whole scan zone; servo 5.1 N·m usable (10.3 N·m stall)	✗ Not OK
5	Axis 4 (wrist) holding torque	2.2 N·m drawn pose, 3.7 N·m worst; 5.1 N·m usable	✓ OK
6	Axis 1 (base) bearing arrangement	18.7–28.0 N·m tilting moment; the single-direction thrust bearing resists only 3.9 N·m, the rest has no load path except the motor shaft	✗ Not OK
7	Axis 1 drive torque	≤ 0.8 N·m at scanning speeds vs 20 N·m holding — but load-to-rotor inertia ratio 938–2,202 : 1	▲ OK torque, poor drive
8	Shafts, keys, bearings (present torque)	Shaft FoS 2.3–2.6, key crushing 23.7 MPa, bearing $s_0 \geq 73$	✓ OK
9	Link stiffness	≈0.6 mm tip deflection in the worst pose	✓ OK
10	Positioning accuracy (RSS estimate)	≈13 mm as designed; ≈5 mm with Option A open loop (gearbox backlash included); ≈1.7 mm with position loops on joint encoders	✗ Not OK → OK
11	Tipping (base not bolted)	FoS 2.3 drawn pose, 1.5 links horizontal, 0.97 with the report's 2.5 kg end-effector	▲ Bolt down
12	Safety on power loss	No brakes; joints go limp and the probe is pressed with up to ≈20 N plus impact	✗ Not OK
13	Autonomy: accuracy in contact	A probe force of 6.7–19.6 N cancels the shoulder's gravity torque (elbow ≈15.7 N, wrist ≈8.4 N), so gear play is crossed while scanning; needs encoder loops on all four joints	▲ Add encoders
14	Autonomy: sensing	Nothing in the design finds the body or senses probe tilt; needs a depth camera (or touch-probing a known phantom), three load cells and video capture of the ultrasound image	✗ Missing
15	Autonomy: probe orientation	With the robot beside a phantom the probe can be held within 15° of normal on 71 % of its upper surface (51 % exactly normal); sideways tilt is impossible with these joints	▲ Plan around

Bottom line

As drawn, the arm cannot hold its own weight anywhere the probe would be during a scan. Not one probe-vertical position in the assumed scan zone (r 400–750 mm from the base axis, z 100–300 mm above the base) is within the usable torque of Axes 2 and 3; even at the absolute catalogue limits only 47 % of it is holdable, with no margin.

The fix does not need new link lengths. Geared drives at Axes 1 and 2, a 3:1 belt at Axis 3, a lighter end-effector (load cells instead of the 6-axis sensor) and a moment-carrying base bearing (Option A, Section 9) make the whole zone holdable with margin and bring the estimated tip error from ≈13 mm to ≈5 mm.

For autonomous scanning (Section 7) it also needs position loops on encoders at all four joints, a way to find the body (depth camera or touch-probing), three load cells for force and tilt, and video capture of the ultrasound image. With the encoder loops the tip error is ≈1.7 mm.

1. Scope, sources and method

This report checks every calculation in the EIR report against the final CAD drawings, rebuilds the load model from the drawings, and adds the checks a reviewer would expect before fabrication (bearings, shafts, stiffness, accuracy, stability, power and safety). Section 3 reviews the report's own numbers; Section 5 redoes the sizing on the final geometry; Section 6 collects the additional calculations separately. Every number comes from one Python script (calc.py, sent with this report): change a mass or a length there and rerun to update the results.

Conventions and assumptions

- g = 9.81 m/s²; 1 kgf·cm = 0.0981 N·m. Joint torques are the torques needed to hold position; positive = lifting the arm.
- Angles: θ_2 measured from horizontal; θ_3, θ_4 relative to the previous link. Drawn pose (measured): θ_2 = 74.1°, θ_3 = −59.9°, θ_4 = −60.4° (the 120° interior angles of the assembly drawing).
- Joint centres measured on the final assembly drawing (sheet 11): shoulder to elbow L_1 = 379 mm, elbow to wrist L_2 = 372 mm, wrist to the probe-holder bend 328.8 mm along Link 3; shoulder axis 275 mm above the base underside. The 400 / 350 / 280 mm in the parts list are the lengths of the bent channels, which overhang the joints; revision A of this report used them as joint spacings (Section 2 compares the results).
- Tool: a 45° bend (probe holder, 135° on sheet 10) on the probe axis, 692 mm from the base axis, then 207 mm from the bend to the probe tip (sensor, U-frame, probe) — both measured on sheet 11.
- Masses from drawing volumes × density; parts without drawings are estimated and flagged (Section 4).
- "Usable" capacity: 70 % of a stepper's holding torque at low speed; 50 % of a hobby servo's stall torque for continuous holding. These are rules of thumb — supply voltage, temperature and driver tuning move them.
- Task: fully autonomous scanning — the robot finds the phantom or body, plans coverage sweeps, brings the probe into contact, regulates the force and follows the plan with no operator. For checking: probe roughly vertical over a zone r = 400–750 mm, z = 100–300 mm (robot on the same table as a phantom); probe speed ≤ 20 mm/s; contact force 5–20 N. Reported probe forces are typically 2.4–12 N, with abdominal examinations reaching ≈30 N or more [1].

2. The design as documented

Architecture

Axis	Motion	Actuator (as listed)	Transmission	Support bearings
1	base yaw	"Servo motor 20 Nm Axis 1" — NEMA-42 frame (inferred)	direct, via "Axis 1 coupler" (no drawing)	SKF 51209 single-direction thrust ball bearing
2	shoulder pitch	"Servo motor 12 Nm Axis 2" — NEMA-42 frame (inferred)	direct: keyed coupler Ø19/Ø15 → Ø15 shaft → keyed nylon connecting block	2 × SKF 16002 in aluminium pillow blocks
3	elbow pitch	DSServo DS5180 "80 kg" servo	direct, via Coupler 3 and Axis-3 shaft	2 × SKF 16002 in nylon bearing blocks
4	wrist pitch	DSServo DS5180 "80 kg" servo	direct, via Coupler 3	2 × SKF 16002 in nylon bearing blocks
(5)	—	"Axis 5 link" in the final parts list, but no fifth actuator	—	—

The report calls the arm five-DOF, but both parts lists contain four actuators; axes 2–4 are parallel, so the arm positions the probe in 3D and sets one tilt angle (in the arm's vertical plane). It cannot roll the probe about its own axis or tilt it sideways independently of the base yaw.

The two stepper axes are NEMA-42 (110 mm frame) motors, inferred from the drawings: the Axis-1 clamping plate has a Ø125.72 mm bolt circle (= 88.9 mm square, the NEMA-42 hole spacing) and a Ø55.52 mm recess "for locating servo head" (= the 2.186 in NEMA-42 pilot); the Axis-2 pillow block carries 4 × M8 on the same 88.9 mm square, and the coupler's keyed Ø19 bore matches a NEMA-42 shaft. The report calls them servo motors and shows a stepper photograph. Typical catalogue data: a 20–21 N·m NEMA-42 is ≈150 mm long and ≈8.4 kg; a 12 N·m one ≈115 mm and ≈6 kg; rotor inertia 10 900 and 7 100 g·cm² respectively; shaft limits 220 N radial (20 mm from the flange) and 60 N axial [6].

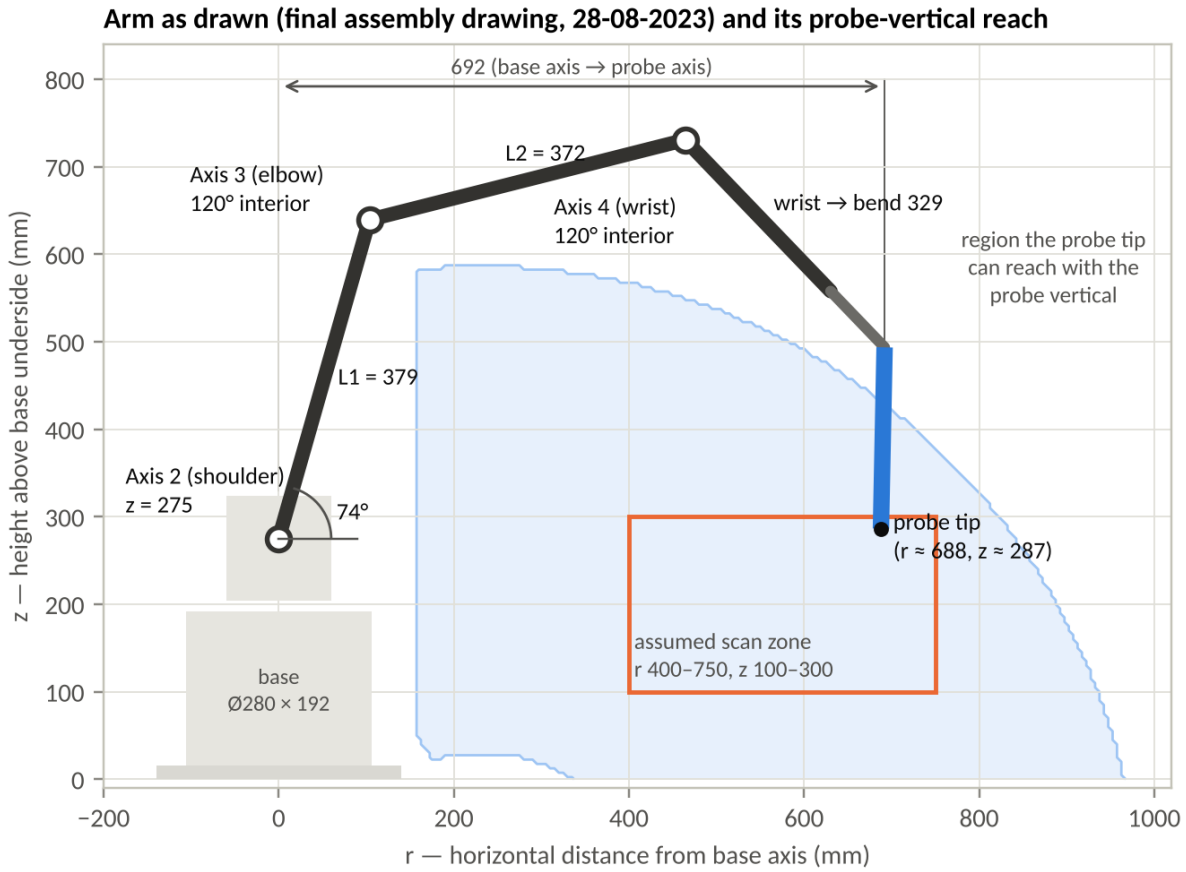


Figure 1. Stick model of the arm in the drawn pose (computed from the assembly dimensions), the region the probe tip can reach with the probe held vertical (joint limits ±90° about the drawn servo angles, links kept above the tip), and the scan zone assumed in this report.

Quantity	Value	Source
Joint spacing, shoulder → elbow / elbow → wrist	379 / 372 mm (channels 400 / 350 mm long)	measured, sheet 11
Wrist → 45° bend on the probe axis / bend → probe tip	328.8 mm along Link 3 (channel 280 long) / 207 mm	measured, sheet 11
Shoulder (Axis 2) height above base underside	275 mm (base stack 16 + 160 + 16 = 192 mm)	assembly + sheets 1–3
Drawn pose	θ2 = 74°, interior angles 120° at elbow and wrist	assembly drawing
Reach, base axis → probe axis	692 mm (model: bend point at 692.1 mm)	measured, sheet 11
Probe tip in the drawn pose (model)	r = 687.9 mm, z = 286.5 mm, probe 1° off vertical	computed
Base footprint	Ø280 mm aluminium plate, 4 × Ø10.5 holes on Ø245 PCD	sheet 1
Links	pairs of 1 mm mild-steel C-channels, 49 × 20 mm	sheet 8 (Link 1)

Revision B: what the measured joint spacings change

Result	Revision A (400 / 350 / 280)	Revision B (379 / 372, measured)	Change
Axis 2, drawn pose / links horizontal (N·m)	13.4 / 25.0	13.6 / 25.0	+2 % / 0 %
Axis 3 over the scan zone (N·m)	8.9–9.8	9.8–10.3	+5 %
Axis 4, drawn pose (N·m)	2.31	2.19	−5 %
Scan zone holdable: as designed (usable) / catalogue limits / Option A	0 / 77 / 100 %	0 / 47 / 100 %	verdict unchanged
Probe tip, drawn pose (r, z in mm)	704, 329	688, 287	matches sheet 11

3. Review of the original torque calculation (report pp. 8–10)

The report sizes one joint ("axis-4", a 250 mm lever on a 330 mm bar) with a motor-sizing tool. Inputs stated: force/torque sensor 1 kg, gripper 1 kg, probe 0.5 kg, arm 1 kg; 60 rpm; 1.5 s acceleration/deceleration; ±2° stopping accuracy; safety factor 1.5; efficiency 75 %. Tool output: J_L = 0.2914 kg·m², T_a = 1.831 N·m, T_L = 9.147 N·m, T = 16.47 N·m (167.95 kgf·cm); electromagnetic brake requested.

3.1 Unit conversion

T = 16.47 N·m ÷ 0.0981 N·m per kgf·cm = **167.9 kgf·cm** ✓

Matches the report. The discussion on p. 10 quotes 164 kg·cm — a minor inconsistency.

3.2 Acceleration torque

$\omega = 2\pi N / 60 = 2\pi \times 60 / 60 = 6.283$ rad/s; $\alpha = \omega / t_a = 6.283 / 1.5 = 4.189$ rad/s²

$J_L \cdot \alpha = 0.2914 \times 4.189 =$ **1.221 N·m**

Report value 1.831 N·m = 1.500 × 1.221 N·m

The report's T_a already contains a factor of 1.5, and the total below multiplies by 1.5 again.

3.3 Total required torque

$T = (T_L + T_a) \times S_f = (9.147 + 1.831) \times 1.5 =$ **16.47 N·m** ✓ (arithmetic reproduces)

With S_f applied once: (9.147 + 1.221) × 1.5 = 15.55 N·m

3.4 Can T_L and J_L be traced to the stated masses?

Payload at the 250 mm lever: 1 + 1 + 0.5 = 2.5 kg; arm 1 kg with its centre of mass at 125 mm (bar 330 mm, pivots 250 mm apart).

$T_{L,static} = (2.5 \times 0.250 + 1.0 \times 0.125) \times 9.81 =$ **7.36 N·m**; ÷ η (0.75) = 9.81 N·m (report: 9.147 N·m)

$J_L = 2.5 \times 0.25^2 + (1.0 \times 0.33^2 / 12 + 1.0 \times 0.125^2) =$ **0.181 kg·m²** (report: 0.2914 kg·m², 61 % higher)

Neither value can be reproduced. The counterweight mentioned on p. 10 has no stated mass or position, and the tool inputs are not shown, so the 16.47 N·m result cannot be audited. The method itself — $T = (T_L + T_a) \times S_f$ — is sound; Section 5 applies it to the final geometry.

3.5 Geometry used vs the final CAD

	Link 1	Link 2	Link 3	Total
Report sketch, member lengths (p. 8)	425 (at 45°)	250 (drawn)	353 (at 45°)	1028
Report bar below the sketch	425	350	250	1025
Parts list, bent-channel lengths	400	350	280	1030
Final assembly, joint centres (measured, used here)	379	372	329 (to bend)	1080

The sketch and the bar under it disagree with each other (Link 2 drawn as 250 but listed as 350; Link 3 drawn as 353 but listed as 250), and both differ from the final CAD. The channel lengths in the parts list are not joint spacings either: the channels overhang the joint centres.

3.6 Motion specification

Sizing speed 60 rpm = 360°/s. A 20 mm/s probe sweep needs at most 4.4 °/s at any joint (Section 6.1) — about 80 times slower.

Stopping accuracy ±2° at the 692 mm reach = ±(2 × π / 180) × 692 = **±24.2 mm** at the probe.

The speed requirement inflates the acceleration term but is irrelevant for scanning; the ±2° accuracy is far too coarse for probe placement. What matters for this robot is holding torque against gravity and fine resolution at low speed.

3.7 Selected actuator vs the report's own result

> DS5180 "80 kg": 80 kgf·cm × 0.0981 = **7.85 N·m**; catalogue stall 85 / 98 / 105 kgf·cm at 6.0 / 7.4 / 8.4 V = 8.34 / 9.61 / 10.30 N·m [5]
> Coverage of the report's 16.47 N·m: 48 % (named rating), 63 % (stall at 8.4 V)
> The servo cannot meet the requirement the report itself computed, and a hobby servo should not be held near stall continuously.

3.8 Statements in the report that need correcting

- "Structural steel … relatively low weight compared to other materials" — steel is ≈2.9 times denser than aluminium. The links are light only because they are 1 mm sheet.
- "Stepper motors are not affected by acceleration … the rate of rotation does not increase as the load is applied" — an open-loop stepper loses steps when load plus inertia torque exceeds its pull-out torque, and it lags its commanded angle under load (Section 6.8).
- "Five degree of freedom robot" — four actuators are listed (Section 2).
- The control scheme (adaptive backstepping position control + PID force control + event triggering) computes joint torques; steppers and hobby servos are position-commanded and give no torque control or position feedback. Force control on this hardware has to be an outer admittance loop that adjusts position set-points.
- References [1]–[3] of the report could not be verified. Reference [1] is described as remote robotic ultrasound with cloud robotics, but RASimAs is an EU FP7 project on a regional-anaesthesia simulator and assistant (CORDIS 610425) [8]. Use verified reviews instead, e.g. [2], [3].
- Probe-holder text refers to "the design shown in figure 4"; Figure 4 is the servo.

Item	Report	Check	Remark
kgf·cm conversion	167.95	167.95	✓ OK
Acceleration torque T_a	1.831 N·m	1.221 N·m	▲ ×1.5 applied twice
Total T	16.47 N·m	16.47 N·m	✓ arithmetic OK
Load torque T_L	9.147 N·m	7.36 (static), 9.81 (÷η)	▲ not traceable
Load inertia J_L	0.2914 kg·m²	0.181 kg·m²	▲ not traceable
Geometry	425/350/250	379/372/329 (measured joint spacing)	▲ superseded
Actuator chosen	80 kgf·cm servo	needs ≥ 16.47 N·m (own result)	✗ inconsistent

4. Mass model built from the drawings

Each drawn part's volume was computed from its dimensions (holes, recesses, keyways and chamfers deducted) and multiplied by density: aluminium 2.70, mild steel 7.85, nylon (PA6) 1.14 g/cm³. Purchased items use catalogue masses. Parts without drawings are estimated and marked "est."; the links without drawings (Link 2, Link 3) are assumed to use the same 1 mm C-channel as Link 1.

Group	Part	Qty	Material	Mass each (kg)	Basis
Base (fixed)	Base plate (Ø280×16)	1	Al	2.623	drawing sheet 1
Base (fixed)	Base motor housing (Ø212/Ø180×160)	1	Al	4.244	drawing sheet 2
Base (fixed)	Axis-1 servo clamping plate (Ø212×16)	1	Al	1.366	drawing sheet 3
Turntable stage	Axis-2 pillow block RHS (motor side)	1	Al	0.743	drawing sheet 4
Turntable stage	Axis-2 pillow block LHS (sheet 5)	1	Al	0.751	drawing sheet 5
Turntable stage	Axis-2 shaft Ø15 g6 × 170	1	MS	0.226	drawing sheet 6
Turntable stage	Axis-2 coupler (Ø19/Ø15 keyed)	1	MS	0.334	drawing sheet 6
Turntable stage	Servo-2 spacer Ø18×84.5	4	MS	0.127	drawing sheet 6
Arm	Axis-2 connecting block (nylon)	1	PA6	0.312	drawing sheet 7
Arm	Link-1 C-channel 1 mm (pair)	2	MS	0.265	drawing sheet 8

Group	Part	Qty	Material	Mass each (kg)	Basis
Arm	Elbow station (servo, housing, block, coupler, 2 bearing blocks, 2 bearings, shafts)	1	mixed	0.842	sheets 8-10 + estimates
Arm	Link-2 channel (pair, assumed same section)	2	MS	0.235	no drawing - assumed
Arm	Wrist station (servo, housing, block, coupler, bearing blocks, bearings, shaft, axis-5 link)	1	mixed	0.765	sheets 8-10 + estimates
Arm	Link-3 channel (pair, assumed same section)	2	MS	0.188	no drawing - assumed
Arm	End effector (probe holder, F/T sensor + bracket, clamp, probe)	1	mixed	0.795	sheet 10 + estimates
Base (fixed)	Axis-1 motor NEMA-42 (20 N·m class)	1	-	8.400	catalogue class
Base (fixed)	Thrust bearing SKF 51209	1	-	0.300	catalogue
Base (fixed)	Fasteners (base)	1	-	0.250	estimate
Turntable stage	Axis-1 coupler / turntable (no drawing)	1	Al	0.750	estimate
Turntable stage	SKF 16002 bearings (Axis 2)	2	-	0.025	catalogue
Turntable stage	Axis-2 motor NEMA-42 (12 N·m class)	1	-	6.000	catalogue class
Turntable stage	Fasteners (yaw stage)	1	-	0.150	estimate

Assembly	Mass (kg)	Comment
Arm beyond the shoulder (realistic end-effector)	4.09	end-effector 0.79 kg: holder 0.12, F/T sensor 0.17, bracket 0.08, clamp 0.12, probe 0.30 (est.)
Arm with the report's end-effector assumption	5.79	end-effector 2.5 kg (sensor 1 + gripper 1 + probe 0.5)
Elbow station / wrist station	0.84 / 0.76	servo 0.158, sheet housing, nylon block, coupler, bearing blocks, bearings, shafts, fasteners
Everything that rotates with Axis 1	13.60	includes the 6 kg Axis-2 motor mounted ≈205 mm off-axis
Whole robot	30.8	base stack 17.2 kg, of which the Axis-1 motor 8.4 kg

5. Corrected joint-torque calculation on the final geometry

5.1 Kinematic model

Planar forward kinematics in the arm's vertical plane (r = horizontal distance from the base axis, z = height above the base underside), with $\theta_{23} = \theta_2 + \theta_3$, $\theta_{234} = \theta_{23} + \theta_4$ and probe direction $\psi = \theta_{234} - 45°$:

$r_T = L_1 \cos\theta_2 + L_2 \cos\theta_{23} + (L_3 + s) \cos\theta_{234} + b \cos\psi$
$z_T = 275 + L_1 \sin\theta_2 + L_2 \sin\theta_{23} + (L_3 + s) \sin\theta_{234} + b \sin\psi$
Base yaw θ_1 rotates this plane: $X = r \cos\theta_1$, $Y = r \sin\theta_1$, $Z = z$
Drawn pose ($\theta_2 = 74.1°$, $\theta_{23} = 14.2°$, $\theta_{234} = -46.2°$, $\psi = -91.2°$): $r_T = 103.6 + 360.6 + 227.8 - 4.2 =$ **687.9 mm**; $z_T = 275 + 364.6 + 91.3 - 237.1 - 207.2 =$ **286.5 mm**

$L_3 + s = 328.8$ mm and $b = 207$ mm are measured on sheet 11: the bend sits on the probe axis, 692 mm from the base axis.

Joint i	θ_i (variable)	d_i (mm)	a_i (mm)	α_i
1 (base yaw)	θ1	275	0	90°
2 (shoulder)	θ2	0	379	0
3 (elbow)	θ3	0	372	0
4 (wrist)	θ4	0	328.8 (to the bend)	0
tool (fixed)	−45° at the bend	0	207 (bend to tip)	0

Standard Denavit–Hartenberg parameters (lateral offsets between the link pairs are neglected).

5.2 Gravity torque — worked calculation for the drawn pose

For each joint j, the holding torque is the sum of the weights of everything beyond it times their horizontal distance from that joint, minus the moment of any upward reaction F_n at the probe tip:

$$\tau_j = \Sigma\, m_i\, g\, (x_i - x_j) - F_n\, (x_T - x_j)$$

Joint positions in the drawn pose: $x_2 = 0$, $x_3 = 103.6$ mm, $x_4 = 464.3$ mm

Item	m (kg)	x (mm)	Moment about Axis 2 (N·m)	about Axis 3	about Axis 4
Connecting block	0.312	6.5	0.020	—	—
Link-1 pair	0.529	57.4	0.298	—	—
Elbow station	0.842	103.6	0.856	—	—
Link-2 pair	0.470	316.9	1.461	0.983	—
Wrist station	0.765	464.3	3.483	2.705	—
Link-3 pair	0.376	591.7	2.182	1.800	0.470
Probe holder	0.125	661.2	0.810	0.683	0.241
F/T sensor + bracket + clamp + probe	0.670	690.0	4.535	3.854	1.483
Holding torque, free space			**13.65**	**10.03**	**2.19**

Example row — wrist station about Axis 2: 0.765 kg × 9.81 × 0.4643 m = 3.48 N·m. The two heaviest contributors are the end-effector and the wrist station, which is why lightening the end-effector matters more than lightening the links.

5.3 Results for all load cases and comparison with the actuators

Load case	Axis 2 (N·m)	Axis 3 (N·m)	Axis 4 (N·m)
Drawn pose, free space, realistic end-effector (0.80 kg)	13.65	10.03	2.19
Drawn pose, free space, report's end-effector (2.5 kg)	25.22	19.87	6.01
Drawn pose, pressing 20 N on the body	-0.11	-1.66	-2.28
Links horizontal, free space, realistic	24.98	11.75	3.67
Links horizontal, free space, report's end-effector	44.42	24.84	10.54
Catalogue rating (holding / stall at 8.4 V)	12.0	10.30	10.30
Usable continuous (70 % / 50 %)	8.40	5.15	5.15

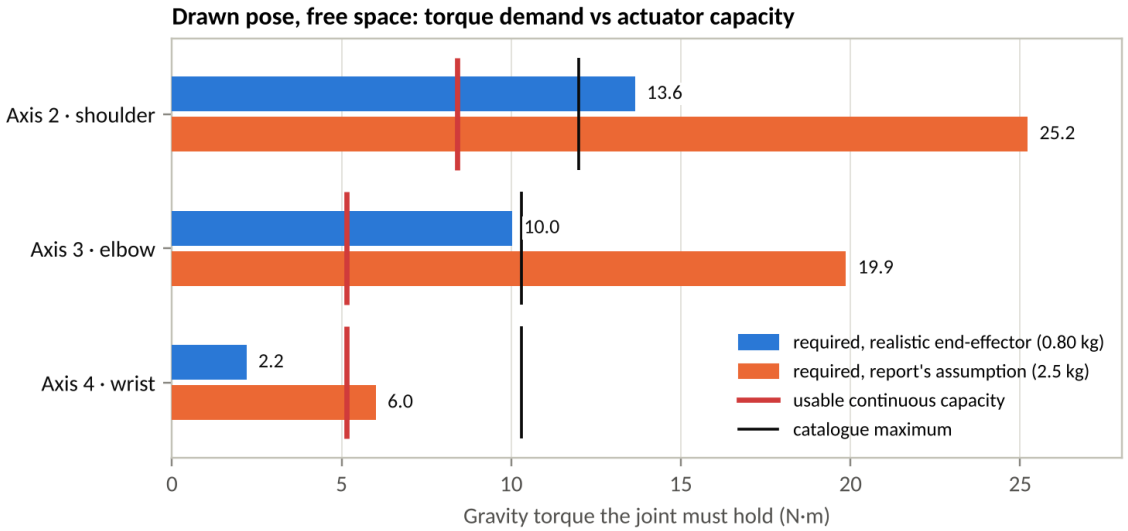


Figure 2. Drawn pose, free space. Axis 2 needs 1.6 times its usable torque and more than its holding rating; Axis 3 needs 1.9 times its usable torque. With the report's 2.5 kg end-effector assumption every pitch axis is short.

5.4 Torque over the whole scan zone

Holding torques were computed at 5 mm spacing for every probe-vertical position the arm can reach (inverse kinematics, elbow up). Over the assumed scan zone, Axis 2 needs 3.2–15.7 N·m, Axis 3 needs 9.8–10.3 N·m and Axis 4 a constant 2.25 N·m (with the probe vertical, Link 3 always sits at −45°).

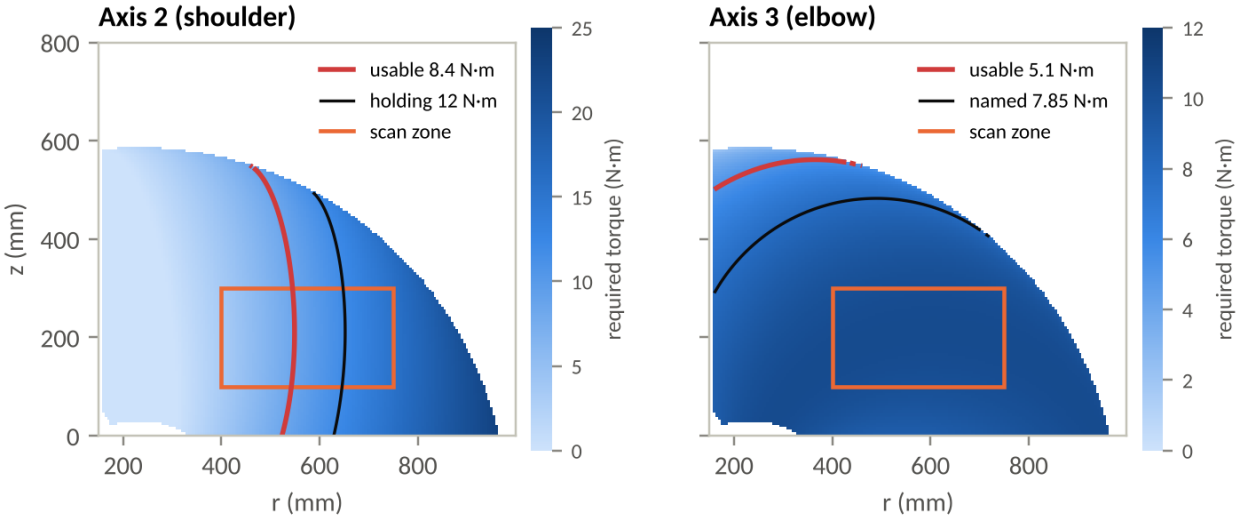


Figure 3. Required holding torque (realistic masses, probe vertical). Red = usable capacity, black = catalogue rating. Axis 3 stays above its usable 5.1 N·m everywhere below z ≈ 505 mm, i.e. everywhere a probe would touch a phantom or patient.

Fraction of the scan zone that can be held	Result
As designed, within usable capacities	**0 %** — the only holdable probe-vertical positions are z ≥ 505 mm and r ≤ 445 mm
As designed, at full catalogue ratings (no margin)	47 % — elbow servo at 95–100 % of stall continuously
Option A (Section 9)	**100 %** — Axis 2 ≤ 14.7 of 38 N·m, Axis 3 ≤ 9.5 of 14.7 N·m

5.5 Effect of probe contact

Pressing on the body is not the problem: the upward reaction at the probe offsets the arm's weight. With 20 N of contact in the drawn pose the joints need only -0.1, -1.7 and -2.3 N·m. The maximum downward force before a joint saturates is ≈26 N as designed (limited by the elbow) and ≈32 N with Option A (wrist-limited) — both above the 20 N design force. The difficulty is free space: approach, retraction and moving between scan lines, where the arm must carry its full weight. For an autonomous scan contact has a second effect: within the normal force range the joint loads pass through zero, so gear play is no longer held closed by gravity (Section 7.2).

> Maximum press force at joint j: $F_{max,j} = (\tau_{g,j} + \tau_{usable,j}) / (x_T - x_j)$
> Elbow, as designed: (10.03 + 5.15) / 0.584 m = **26.0 N**
> "Limp" force if a joint loses torque: $F = \tau_{g,j} / (x_T - x_j)$ → Axis 2: 19.8 N, Axis 3: 17.2 N, Axis 4: 9.8 N (static, before any impact)

5.6 Dynamic torques — the report's motion profile vs scanning

Axis and pose	Inertia (kg·m²)	Report profile α = 4.19 rad/s² (N·m)	Scanning α = 0.35 rad/s² (N·m)
Axis 1, drawn pose	1.02	4.28	0.36
Axis 1, links horizontal	2.40	10.06	0.84
Axis 2, drawn pose	1.17	4.91	0.41
Axis 2, links horizontal	2.12	8.86	0.74
Axis 3, drawn pose	0.54	2.26	0.19

Inertias include the links' own inertia and, for Axis 1, the Axis-2 motor offset (≈0.29 kg·m² of the turntable stage alone). Scanning profile: 10°/s reached in 0.5 s. At scanning speeds the inertial torque is under 1 N·m everywhere — gravity dominates the sizing.

Axis 1 with the report's own method: $T = J\alpha / \eta \times S_f = 2.40 \times 4.19 / 0.75 \times 1.5 =$ **20.1 N·m** (links horizontal)

This matches the 20 N·m base motor, which suggests how it was chosen. At scanning speeds the base needs under 1 N·m; what it really needs is a stiff, moment-carrying bearing and a better inertia match (Sections 6.3 and 6.8).

6. Additional calculations

These checks are not in the report but are needed before fabrication. Each uses the mass model of Section 4.

6.1 Jacobian and joint speeds for scanning

Velocity mapping $(v_r, v_z, \omega_\psi) = J(\theta) \cdot (\omega_2, \omega_3, \omega_4)$, with J in mm/rad for the drawn pose:
J = [-11.5 353.0 444.3 ; 687.9 584.3 223.6 ; 1 1 1], det J = -122,013 (far from singular)
20 mm/s radial sweep, probe angle fixed: $(\omega_2, \omega_3, \omega_4) = (-3.39, 4.36, -0.97)$ °/s
20 mm/s vertical approach: (-0.86, 4.28, -3.42) °/s; 20 mm/s lateral sweep by base yaw: $\omega_1 = 1.67$ °/s

Every joint moves at under 4.4 °/s. Actuator speed is never limiting; smoothness at very low speed is (Sections 6.8–6.9).

6.2 Reach and placement

With the probe vertical, the tip can reach r = 160–885 mm at z = 200 mm, and the whole assumed scan zone is reachable (100 %). Reach is therefore not a constraint; place the base so the scan area sits 400–750 mm from its axis and 100–300 mm above its underside. The base yaw sweeps the scan plane sideways, so a lateral move also rotates the probe's image plane (±15° for ±150 mm at 550 mm) — a consequence of having no probe-roll axis.

6.3 Axis 1 bearing: tilting moment

Weight on the thrust bearing W = 13.60 kg × 9.81 = 133.4 N; ball pitch radius of SKF 51209 = (45 + 73)/4 = 29.5 mm
Moment the thrust bearing can resist before one side lifts: $M = W \cdot r_p = 133.4 \times 0.0295 =$ **3.94 N·m**
Applied, drawn pose: arm 13.65 N·m (in plane) and Axis-2 motor offset 12.73 N·m (sideways) → resultant $\sqrt{(13.65^2 + 12.73^2)}$ = **18.7 N·m** (4.7 × the lift-off moment)
Links horizontal: 28.0 N·m (7.1 ×); with the report's end-effector 46.2 N·m

The excess has no designed load path except the motor shaft. For comparison, the NEMA-42 limit of 220 N radial at 20 mm from the flange is ≈4.4 N·m — so the base will wobble and the motor bearings will be overloaded. Axially, the bearing must carry all 133 N; if the coupler lets weight reach the motor shaft it exceeds the 60 N axial limit.

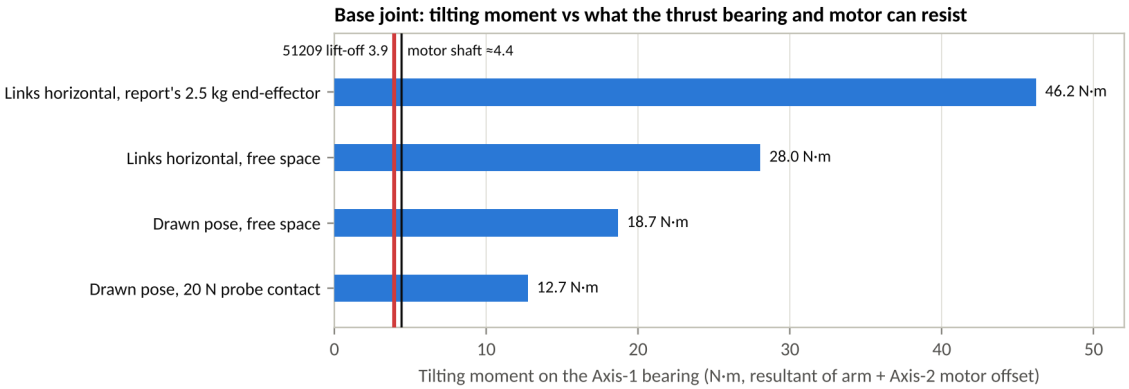


Figure 4. Tilting moment on the base joint for four load cases against what the thrust bearing (red) and the motor shaft (black) can resist.

Fix: carry the turntable on a bearing set that takes moment — a crossed-roller or four-point slewing bearing, or a vertical shaft in two angular-contact bearings (e.g. 2 × 7009 back-to-back, ≈50 mm apart: radial reaction ≈ 28 N·m / 0.05 m ≈ 561 N, trivial for that bearing) — and drive it through a flexible coupling so the motor carries torque only.

6.4 Tipping stability

Case	Overturning moment (N·m)	CG offset e (mm)	FoS = 140 mm / e
Drawn pose, free space	18.7	62	2.27
Drawn pose, 20 N contact	12.7	42	3.32
Links horizontal, free space	28.0	93	1.51
Links horizontal, report's end-effector	46.2	145	0.97

$e = M / (m_{total}\ g)$ with m_{total} = 30.8 kg. A free-standing base is marginal at full reach and tips with the report's heavier end-effector. Bolt it down through the 4 × Ø10.5 holes on the Ø245 PCD (M10) or clamp it to the bench.

6.5 Axis-2 shaft, key and hub

Bending: arm weight + 20 N contact = 60.1 N at the connecting block, bearings ≈114 mm apart (load 74/40 mm from them): M = F a b / L = 1.56 N·m

Stresses with first-iteration concentration factors [4]: keyway K_t = 2.14, K_{ts} = 3.0; circlip groove (Ø14.3) K_t = 5.0, K_{ts} = 3.0 $\sigma_{vm} = \sqrt{(\sigma^2 + 3\tau^2)}$, mild steel S_y = 250 MPa. At T = 12 N·m (the motor's maximum): keyway 95 MPa → FoS 2.64; groove 109 MPa → FoS **2.30**

At T = 25.0 N·m (worst gravity torque, which the shaft must carry once a reducer is fitted): keyway FoS 1.27, groove FoS **1.11**

Key 6 mm × 22.5 mm in the coupler at 12 N·m: shear 2T/(d b l) = 11.9 MPa, crushing 2T/(d (h/2) l) = 23.7 MPa

Nylon connecting-block keyway (3 mm × 70 mm): 7.6 MPa at 12 N·m, 15.9 MPa at 25.0 N·m

Torsional twist over ≈60 mm of shaft at the drawn-pose torque: θ = T L / (G J) = 0.119° → 1.4 mm at the probe

Fine as designed. With a reducer the shaft needs EN19/EN24 (S_y ≈ 650 MPa → groove FoS 2.9) or Ø20 (FoS 3.0, but then 16004 bearings). Nylon creeps under sustained load — replace the keyed nylon hub with an aluminium or steel clamp hub.

6.6 Bearings and fits

SKF 16002 (C = 5.85 kN, C_0 = 2.85 kN [7]): worst radial load 39.0 N → $s_0 = C_0/P_0$ = **73**; $L_{10} = (C/P)^3 \approx 3.4 \times 10^6$ million revolutions

SKF 51209 (C = 39 kN, C_0 = 86.5 kN [7]): axial load 133 N → s_0 = **648** — capacity is not the issue, moment is (6.3)

Inner-ring fit: bore 15 (0/−8 µm) on shaft g6 (−6/−17 µm) → −2 to +17 µm, mostly clearance → ring creep and play. Use k5 (+1/+9 µm).

Outer-ring fit: seat Ø32 +0.03/+0.08 on ring 0/−11 µm → 30–91 µm clearance → radial play. Use H7 (0/+25 µm).

Load capacity has huge margin everywhere. The fits, not the bearing sizes, decide the joint play.

6.7 Link stiffness

C-channel 49 × 20 × 1 mm (centre-line 48 × 19.5): $I = t\,h^3/12 + 2\,b\,t\,(h/2)^2$ = 9 216 + 22 464 = 31,680 mm⁴; pair 63,360 mm⁴; EI = 12,672 N·m²

Link 1 as a cantilever, links horizontal (worst): tip deflection 0.103 mm, end slope 0.49 mrad; bending stress at the root 9.7 MPa

Links 1 + 2 + 3 combined, including slope amplification: **≈0.56 mm** at the probe

In-plane stiffness is ample. The thin open channels are weak in torsion on their own; they rely on the end blocks to act as a pair, so keep the bolted end connections tight and add one or two tie plates along each link.

6.8 Stepper behaviour under gravity load

A hybrid stepper (50 rotor teeth, 1.8° step) under a steady load lags its commanded angle by $\Delta\theta = (2 \times 1.8° / \pi) \cdot \arcsin(T_{load} / T_{hold})$
At 70 % of holding torque: Δθ = 0.89° → **10.7 mm** at the probe for direct drive; above 100 % it slips poles and the arm drops
Load-to-rotor inertia ratio, direct drive: Axis 1 938–2,202 : 1, Axis 2 1,652–2,980 : 1 (common guidance: ≤ 10 : 1 open loop, ≤ 30 : 1 closed loop)
Gear ratio for ≤ 30 : 1 at full reach: N ≥ √(ratio / 30) = 8.6 (Axis 1), 10.0 (Axis 2); with 10 : 1 the drawn-pose ratios become 9.4 and 16.5 : 1
Holding stiffness $k = T_{hold} \times 50$ = 600 N·m/rad (Axis 2): first mode ≈ √(k/J)/2π = 3.6 Hz direct, 35 Hz with 10 : 1 (Axis 1: 5.0 → 47 Hz)

Direct drive gives a soft, slowly-settling, under-resolved joint. A reduction fixes resolution, inertia matching and stiffness together; with the Option A drive (NEMA-34, 10:1) the Axis-2 droop is 0.23 mm at the probe.

6.9 Servo command resolution

DS5180: 500–2500 µs for 180° → 0.09 °/µs; dead band 3 µs [5] → **0.27°** (3.2 mm at the probe through the elbow)
PCA9685 driver at a 50 Hz frame: 20 000 µs / 4096 = 4.88 µs per count → 0.44° per count
Same driver at a 333 Hz frame (the DS5180 accepts 50–333 Hz [5]): 0.73 µs per count → 0.066° per count

Use ≥ 330 Hz frames or a microcontroller timer with ≤ 1 µs resolution; a 50 Hz frame alone adds several millimetres of quantisation.

6.10 Positioning error budget

Source (drawn pose)	As designed (mm)	Option A, open loop	Option A, encoder loops	Basis
Axis-1 microstep error (lateral)	1.2	0.0	0.0	±0.1° at the motor; ÷50 through the A1 gearmotor
Axis-1 gearbox backlash (lateral)	0.0	2.0	0.0	±½ of 15–25 arcmin (precision planetary, assumed); mid value shown
Axis-2 stepper load droop	10.7	0.2	0.0	70 % load direct; NEMA-34 + 10:1 at ≈16 % load
Axis-2 key / coupler play	2.8	0.2	0.0	≈0.23° from bore +0.02/+0.05 and key clearance; clamp hubs
Axis-2 gearbox backlash	0.0	2.2	0.0	±½ of 18–25 arcmin (10:1 planetary listing); mid value shown
Axis-3 servo dead band + backlash	4.6	1.5	0.0	±(0.27° + 0.5° backlash, assumed)/2; ÷3 with belt
Axis-4 servo dead band + backlash	3.3	3.3	0.0	same servo, direct drive in both cases
PWM resolution (servos)	3.2	0.5	0.0	50 Hz PCA9685 vs 333 Hz frame
Joint-encoder loops, Axes 1–2	0.0	0.0	0.8	AS5048A after the gearboxes, ±0.05° after calibration; replaces the rows above
Joint-encoder loops, Axes 3–4	0.0	0.0	1.4	±0.05° at the elbow (through 3:1); ±0.14° at the wrist (servo dead band limits the loop)
Shaft twist + link bending	1.5	1.5	0.6	Sections 6.5 and 6.7; with loops only link bending remains (the Axis-2 encoder sits after the shaft)
Bearing fit play (wobble)	0.6	0.1	0.1	Section 6.6 fits vs k5/H7
Root-sum-square	**13.0**	**5.0**	**1.7**	open loop 4.7–5.4 mm over the backlash range

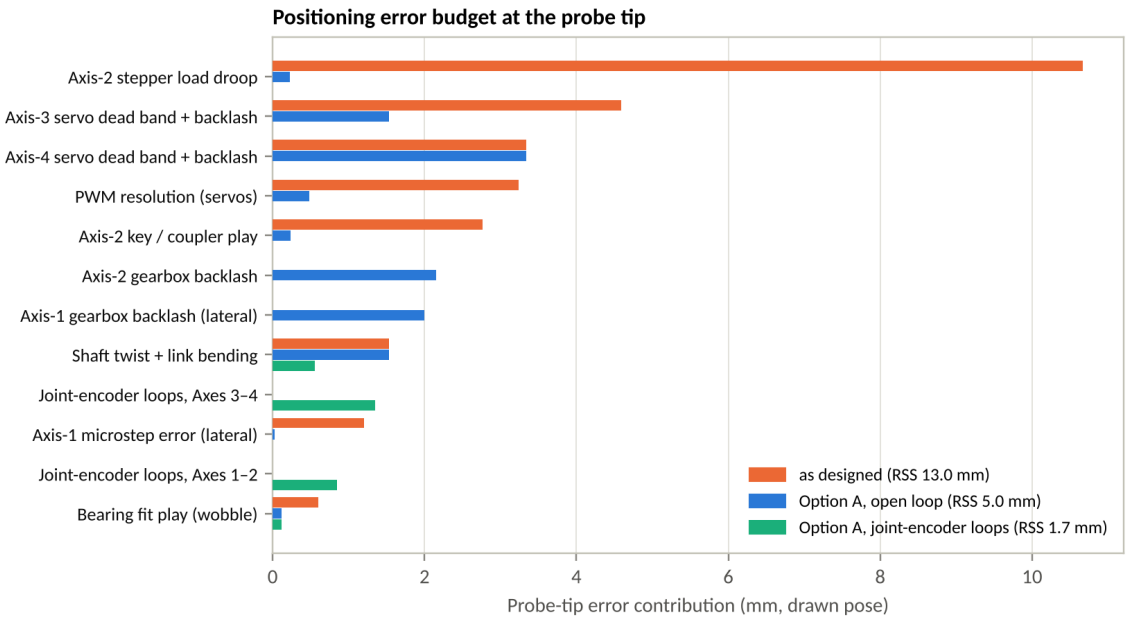


Figure 5. Contributions to probe-tip position error. Direct-drive droop at Axis 2 dominates the as-designed estimate; with Option A open loop the wrist servo and the gearbox backlash dominate; with position loops on the joint encoders the encoder accuracy and link bending remain.

6.11 Contact-force regulation and a series spring

With position-commanded joints, force is regulated by moving the probe into the tissue. The smallest commandable motion along the probe axis (elbow servo dead band) is 2.8 mm; times the contact stiffness this gives the force step. Tissue stiffness is assumed 0.5–2 N/mm for illustration. A soft spring in series lowers the effective stiffness $k_{eff} = k_t k_s / (k_t + k_s)$:

Tissue k_t (N/mm)	No spring	Spring 1.0 N/mm	Spring 0.5 N/mm	0.5 N/mm + 3:1 elbow
0.5	1.4 N	0.9 N	0.7 N	0.23 N
1.0	2.8 N	1.4 N	0.9 N	0.31 N
2.0	5.5 N	1.8 N	1.1 N	0.37 N

A 0.5–1 N/mm spring with 20–40 mm travel (to reach 20 N) between the wrist and the probe clamp, read by load cells, gives ≈1 N force resolution, absorbs breathing motion and limits impact force. Three load cells instead of one also give the two tilt moments an autonomous robot needs to keep the probe normal (Section 7.5). This is cheaper and more useful here than a 6-axis sensor.

6.12 Power budget

Load	Estimate (W)	Basis
Axis-1 stepper, standstill copper loss	68	$2 \times I^2 R = 2 \times 6.5^2 \times 0.8\ \Omega$ (21 N·m class)
Axis-2 stepper, standstill copper loss	35	$2 \times 6.0^2 \times 0.48\ \Omega$ (12 N·m class)
Two servos holding (≈3 A each at 7.4 V)	44	peak 2 × 6.5 A × 8.4 V ≈ 109 W at stall
Electronics	10	microcontroller, encoders, load-cell amplifier
Continuous / peak	**177 / 283**	stepper drivers +20 % (continuous), +60 % (moving)

Size the stepper supply to the chosen drivers (NEMA-42 drivers often take high-voltage AC or 60–110 V DC), a separate 7.4 V ≥ 15 A supply for the servos, and a 5 V logic rail. Servo current spikes must not share the logic supply.

6.13 Safety on power loss and E-stop

- Steppers lose holding torque when de-energised and hobby servos go limp: the arm falls onto the patient or phantom, pressing with ≈20 N statically, plus impact. The sizing tool's "electromagnetic brake" requirement was not carried into the parts list.
- Fit power-off (spring-applied) brakes on Axis 2 and, with a belt drive, on Axis 3 — or counterbalance the arm so it is near neutral when unpowered.

- Make the E-stop a controlled stop that holds position and then engages brakes; a bare power cut is itself a hazard on this arm.
- Software force limit (≈25 N, above the 20 N design force), soft joint limits, and a mechanical breakaway or spring end-stop at the probe.

7. Autonomous operation — extra checks

The BTP robot scans on its own: it finds the phantom or body, plans coverage sweeps, brings the probe into contact, regulates the force and follows the plan with nobody correcting it. A teleoperated arm can lean on the operator's eyes and hands; an autonomous one has to sense and correct everything itself. This section checks what that adds to Sections 5 and 6.

7.1 What the robot has to do alone

Task step	What the robot must do	Sensing needed	As designed	Option A + A11–A12
Find the body	build a surface model (height map and normals)	depth camera, or touching known points	✗ none	✓ camera or touch-probing
Plan the scan	coverage sweeps within reach, torque and orientation limits	robot model + surface model	● —	✓ URDF + MoveIt 2; 5.4, 7.5
Approach and contact	slow approach until contact, then hold 5–20 N	probe force	✓ 6-axis sensor	✓ three load cells + spring
Keep the probe normal	tilt the probe to the skin	surface normals, tilt moments	▲ no joint feedback	▲ in-plane only (7.5)
Follow the path	≈±5 mm at the tip without visual correction	joint angles after the gearboxes	✗ none	✓ encoders on all joints
Check the image	record the image with the probe pose	ultrasound video into the PC	✗ none	✓ video capture
Stay safe unattended	limit force, stop safely, hold on power loss	force, watchdog, brakes	✗ none	✓ A9

7.2 Contact force unloads the joints

With the probe vertical, a contact force F along the probe pushes back on every joint with a moment F ($x_T - x_j$) that opposes the gravity torque. The force that cancels the gravity torque of joint j is

$F_{0,j} = \tau_{g,j} / (x_T - x_j)$ (Option A end-effector, probe vertical, every point of the scan zone)
Axis 2 (shoulder): F_0 = 6.7–19.6 N, median 15.0 N
Axis 3 (elbow): F_0 ≈ 15.7 N everywhere in the zone; Axis 4 (wrist): F_0 = 8.4 N (Link 3 always at −45°)

These forces lie inside the 5–20 N used for scanning. Near them the gear teeth are unloaded and the play between them is free: a small disturbance moves the probe through the whole backlash, and as the force controller presses harder or eases off, the load changes side. An operator corrects this by eye; an autonomous robot has to measure the joint angles after the play and correct them itself.

7.3 Gearbox backlash and joint-encoder loops

Base, 50:1 precision planetary, 15–25 arcmin: ±½ backlash at 688 mm → ±1.5–2.5 mm sideways at the tip
Shoulder, 10:1 planetary, 18–25 arcmin: ±1.8–2.5 mm at the tip
Option A open loop, all sources (Section 6.10): **4.7–5.4 mm** RSS — at the edge of the ±5 mm target
Position loops on absolute encoders after the gearboxes, all four joints (±0.05° at Axes 1–3, ±0.14° at the wrist): **1.7 mm**
Lean build, economy 47:1 base gearbox (backlash up to 3°): ±18 mm sideways without a base encoder (RSS 18 mm); with AS5600 loops (±0.1°) on all four joints: **2.5 mm**

The base axis is occupied by the drive, so its encoder reads the turntable through a 1:1 timing belt or gear pair. On the stepper axes the loop is closed in software (read the encoder, correct the step count, approach targets from one side); on the servo axes the PWM command is trimmed until the encoder reads the target. Calibrate each magnetic encoder once against a reference to reach the stated accuracy.

7.4 Finding the body surface

- **Depth camera (A11).** An RGB-D camera on a stand 0.6–0.8 m above the scan area gives a point cloud, from which the planner builds a height map and surface normals. Stereo depth error grows with distance (the Orbbec Gemini 2 is specified at ≤ 2 % at 2 m [9]), so plan on the camera map but finish every approach on force: move slowly along the probe axis until the load cells see contact — the 20–40 mm spring travel absorbs a few millimetres of map error — then switch to force control.
- **Hand-eye calibration.** A printed ChArUco board, calibrated once with the standard OpenCV / ROS 2 tools, puts the camera map in the robot's frame.
- **Without a camera (lean build).** If the phantom's shape is known, touch it at ≈25 points: approach 30 mm at 5 mm/s, retract at 20 mm/s, ≈3 s to the next point → 10.5 s per point, ≈4.4 min in all; then fit the known shape to the points. This works for a rigid phantom, not for a breathing patient.

7.5 Keeping the probe normal to the skin

Joints 2–4 tilt the probe only in the arm's vertical plane; the sideways part of the surface normal cannot be matched. Where the robot stands decides how much that matters. Check: a phantom 300 mm wide, 120 mm high and 300 mm long (elliptic cylinder), its upper surface up to 60° from horizontal, centred 575 mm from the base axis. For every surface point the probe is tilted in the arm's plane to match the normal, and joint limits, link clearance and the Option A torques are checked.

Robot position	Sideways tilt ≤ 10°	Reachable, probe exactly normal	Reachable within 15° of normal	Worst sideways tilt
Beside the phantom (arm plane across the body)	96 %	51 %	71 %	16°
At its end (arm plane along the body)	21 %	100 %	100 %	58°

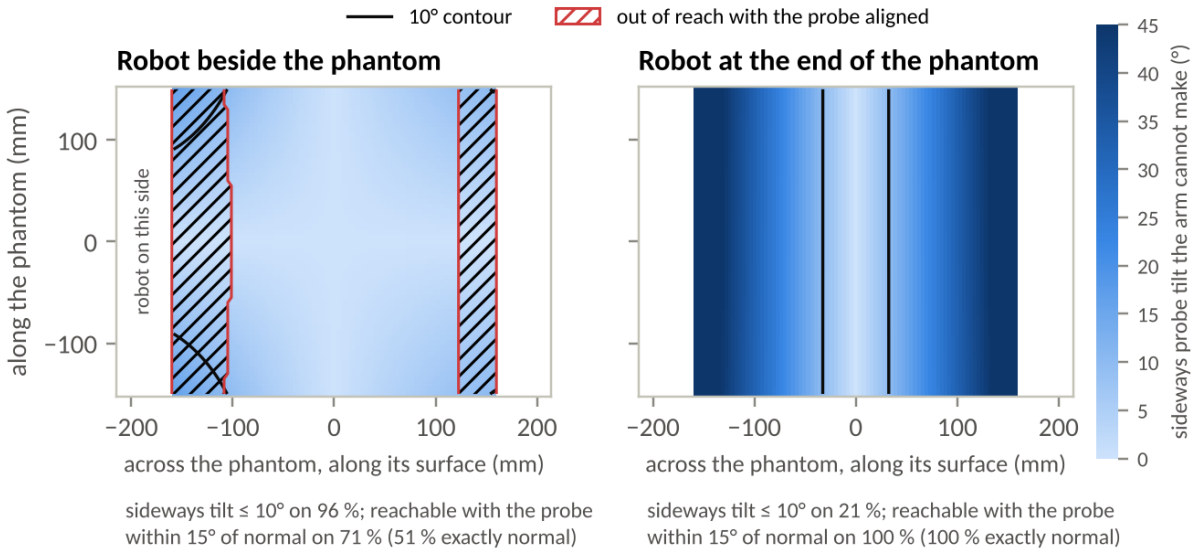


Figure 7. Sideways probe tilt the arm cannot make, over the phantom's upper surface, and the parts that cannot be reached with the probe aligned (hatched). Left: robot beside the phantom (on the left of the plot). Right: robot beyond the bottom edge.

Put the robot beside the phantom or patient, so that its vertical plane runs across the body and the base yaw moves along it: then 96 % of the upper surface needs no more than 10° of sideways tilt. Reach then limits the in-plane tilt instead — the probe can lean about 25–30° toward or away from the robot — so the steep flanks on both sides are out of reach with the probe normal. Covering them needs the robot moved to the other side, the phantom turned between passes, or a sideways-tilt joint at the wrist (Option B).

Three load cells on a circle of radius r under the probe holder measure the force and both tilt moments: $F_i = F_z/3 + 2\,(M_x\,y_i - M_y\,x_i) / (3\,r^2)$
A 10 N contact whose centre sits 3 mm off the probe axis (r = 25 mm): $\Delta F = 2\,F\,e / (3\,r)$ = **0.8 N** between cells

Easily resolved by an HX711 on each cell. The controller tilts the probe in the arm's plane until the in-plane moment vanishes; the sideways moment, which the arm cannot correct, flags poor contact so the planner can skip or revisit that spot.

7.6 Image feedback

A USB video capture of the scanner's output (A12) puts every ultrasound frame on the PC next to the robot's pose. The planner can then log images with their positions, detect lost coupling (a dark image) and repeat a sweep — the minimum needed to close the loop from image to motion later.

8. Drawing errata and missing information

#	Where	Issue	Correction
1	Sheet 5	Titled "Axis-2 Pillow block RHS" but it has no motor holes — it is the mirror/LHS block	Retitle "Axis-2 pillow block LHS (MIR)"
2	Sheets 7, 9, 10	Thread "M6x0.7" is not an ISO metric thread (M6 is 1.0 coarse or 0.75 fine)	M6 × 1.0; use threaded/heat-set inserts in nylon
3	Sheet 6, Axis-2 shaft	Ø15 g6 at the bearing seats gives −2/+17 µm on the 16002 bore (clearance)	k5 at bearing seats
4	Sheets 4, 5, 9	Bearing seat Ø32 +0.03/+0.08 gives 30–91 µm clearance on the outer ring	H7 (Ø32 0/+0.025)
5	Sheet 6, coupler	Bores +0.02/+0.05 with a key → ≈0.2° backlash (≈2.8 mm at the probe)	Clamp hubs, or P9/N9 keyways
6	Sheet 6, keyway	6 mm key on a Ø15 shaft; DIN 6885 uses 5 × 5 for 12–17 mm shafts	Confirm key size and depth
7	Sheet 11 / parts lists	"Rototic Arm Assembly", "clampimg"	Typos
8	Parts lists	Quantities differ between the 16-08 and 28-08 revisions (SKF 16002: 3 + 1 vs 4 + 2; Axis-3 link 2 vs 1); sheet 9 says 4 bearing blocks	Reconcile — see the component-list PDF
9	Sheet 8 vs assembly	Link-1 hole patterns sit ≈30 mm from each end of a 400 mm part; on the assembly the joints are 379 mm apart	Resolved in revision B: 379 / 372 mm measured and used
10	Report p. 7	Probe holder "shown in figure 4" — Figure 4 is the servo	Fix figure reference

Parts with no manufacturing drawing

Axis-1 coupler / turntable · Link 2 (and mirror) · Link 3 (and mirror) · Axis-3 shaft · Axis-3 link · Coupler 3 · Bearing block 3 (if it differs from Bearing block 2) · Axis-5 link · Probe holder 2 · Force-sensor bracket · Probe and bracket assembly · "ID_1" (imported part). The report's statement that the drawings are ready for fabrication (MRL) does not yet hold; about half of the made parts have no drawing.

9. Recommendations

Option A — minimum change, same link geometry

#	Change	Why	Effect
A1	Axis 1: crossed-roller bearing (RU66 class) + flexible coupling; geared NEMA-23 with a 50:1 high-precision planetary in place of the NEMA-42	28 N·m tilting moment vs 3.9 N·m; inertia ratio up to 2,202 : 1	stiff base; inertia ratio 14–32 : 1; ≈40 N·m available
A2	Axis 2: NEMA-34 (8.5 N·m) + 10:1 planetary in place of the NEMA-42, driver current set to ≈38 N·m output, 24 V power-off brake	needs up to 25 N·m vs 8.4 usable	capacity 38 N·m; droop 0.23 mm; inertia ratio 43–78 : 1
A3	Axis-2 shaft in EN19/EN24 (or Ø20 with 16004 bearings); k5/H7 fits; clamp hub instead of keyed nylon hub	groove FoS 1.11 at 25 N·m in mild steel	FoS 2.9; play removed
A4	Axis 3: 3:1 timing-belt reduction from the DS5180; aluminium bearing blocks	needs 9.8–10.3 N·m vs 5.1 usable	capacity 14.7 N·m; resolution ×3
A5	End-effector: three load cells on a 0.5–1 N/mm series spring stage instead of the 6-axis sensor; light clamp (≤ 0.65 kg total)	end-effector is the largest single moment; autonomy needs force and tilt	−1.0 N·m at Axis 2; ≈1 N force resolution; tilt moments for normal alignment
A6	Absolute encoders (AS5048-class) on all four joints, after the gearboxes (base via a 1:1 belt); position loops closed on them; homing switches on Axes 1–2	contact force unloads the joints, so play is crossed while scanning (7.2)	tip error ≈1.7 mm
A7	Servo PWM at ≥ 330 Hz or ≤ 1 µs timers	50 Hz PCA9685 = 0.44° per count	quantisation ≈0.07°
A8	Bolt the base (4 × M10 on Ø245 PCD)	tipping FoS 1.5 at full reach	no tipping
A9	Safety: brakes, hold-then-brake E-stop, 25 N force limit, soft limits	arm falls on power loss	fail-safe
A10	Optional Axis 5: small probe-roll servo (20–35 kgf·cm)	no roll DOF for transverse/longitudinal views	5 DOF as the report intended
A11	Surface sensing: depth camera over the scan area with hand-eye calibration — or touch-probing a phantom of known shape	the robot must find the body and its normals itself	surface model for coverage planning (7.4)
A12	Video capture of the ultrasound image into the PC	image check and recording during autonomous sweeps	every frame logged with the probe pose

Option B — alternatives worth costing

- Counterbalance springs or gas struts at Axes 2 and 3 sized for ≈10 N·m and ≈5 N·m in the mid-scan pose. Smaller motors, and the arm stays put when unpowered; costs some design time for the spring geometry.
- A sideways-tilt joint at the wrist (making the arm 5–6 DOF) if the flanks of a curved abdomen must be scanned with the probe normal (Section 7.5).
- Aluminium links (2 mm sheet or extrusion) to remove ≈0.4 kg from the arm; only worthwhile after A1–A5.

Suggested order of work

- Build a URDF of the arm with the masses of Section 4 and run the scan paths in ROS 2 / MoveIt with joint-torque limits — this checks every change before cutting metal and is also the path-planning test bed.
- Freeze the base bearing (A1) and the Axis-2 drive (A2–A3); they change the most drawings.
- Then the elbow belt (A4), end-effector (A5), sensors and electronics (A6–A7), safety (A9).
- Autonomy layer: camera and hand-eye calibration (A11) or a touch-probing routine, the surface model, coverage planning on it, then force-controlled approach and sweeps on the phantom with image capture (A12).
- Fabricate, then validate on a phantom: holding torque margins, repeatability, force regulation, E-stop behaviour.

References

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[2] F. von Haxthausen, S. Böttger, D. Wulff, J. Hagenah, V. García-Vázquez, S. Ipsen, “Medical robotics for ultrasound imaging: current systems and future trends,” Current Robotics Reports, vol. 2, pp. 55–71, 2021.

[3] Z. Jiang, S. E. Salcudean, N. Navab, “Robotic ultrasound imaging: state-of-the-art and future perspectives,” arXiv:2307.05545, 2023.

[4] R. G. Budynas, J. K. Nisbett, Shigley’s Mechanical Engineering Design — Table 7-1, first-iteration estimates of stress-concentration factors.

[5] DSServo DS5180 SSG data: servodatabase.com/servo/dsservo/ds5180-ssg; Probots listing (torque 85/98/105 kgf·cm, 158 g, 50–333 Hz).

[6] JKongmotor JK110HS NEMA-42 stepper table (holding torque, length, rotor inertia, mass; 220 N radial / 60 N axial shaft limits).

[7] SKF 16002 (C 5.85 kN, C0 2.85 kN) and SKF 51209 (C 39 kN, C0 86.5 kN) catalogue data, via qualitybearingsonline.ca.

[8] CORDIS, “RASimAs — Regional Anaesthesia Simulator and Assistant,” FP7-ICT project 610425, 2013–2016.

[9] Orbbec Gemini 2 / 2L depth camera specification (depth accuracy ≤ 2 % at 2 m), ThinkRobotics listing, seen 13 Oct 2026.

Appendix A — assumption register

Assumption	Value used	If wrong
Bend to probe tip	207 mm (measured, sheet 11)	a longer stack raises Axis 3–4 torque slightly; reach shifts down
End-effector mass (realistic)	0.79 kg	every 0.1 kg at the tip adds ≈0.7 N·m at Axis 2 and ≈0.6 N·m at Axis 3
Link 2 / Link 3 section	same as Link 1 (1 mm MS C-channel)	heavier sections raise Axis 2 and 3 torque
Turntable (Axis-1 coupler)	0.75 kg Al	affects base inertia and tipping only
Usable fractions	70 % stepper holding, 50 % servo stall	higher fractions shrink but do not remove the shortfall (Section 5.4)
Servo backlash	0.5°	error budget term for Axes 3–4
Scan zone	r 400–750 mm, z 100–300 mm	moving the zone closer to the base lowers Axis-2 torque; Axis 3 barely changes
Tissue stiffness	0.5–2 N/mm	force-resolution table only
Mild steel yield	250 MPa (IS 2062 E250)	EN8 gives more margin
Gearbox backlash (Option A)	base 15–25, shoulder 18–25	open-loop error budget (6.10, 7.3)
Joint-loop accuracy	±0.05° (AS5048A, calibrated), ±0.14° at the wrist, ±0.1° (AS5600)	error with encoder loops (6.10, 7.3)
Phantom model	300 × 120 × 300 mm elliptic cylinder, 575 mm from the base axis	coverage fractions in 7.5
Touch-probing	25 points, 5 mm/s approach	time estimate in 7.4